

Leaf Scanner AI Module

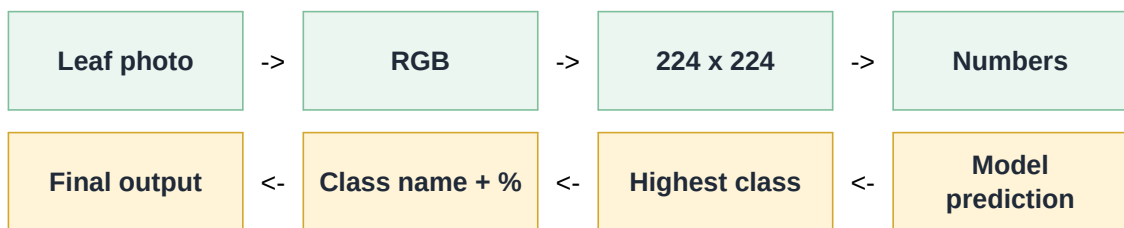
Simple workflow explanation for the Smart Farmer project

Purpose: The module checks a leaf photo and predicts the plant, disease status and confidence percentage.

Complete workflow

1. Select image	The user uploads a leaf photo from the gallery or captures it using the camera/webcam.
2. Convert to RGB	The image is converted to standard Red, Green and Blue colour format.
3. Resize image	The image is resized to 224 x 224 pixels because the MobileNetV2 model expects this size.
4. Convert to numbers	The image pixels are converted into a NumPy numerical array. A deep-learning model understands numbers, not the photo directly.
5. Predict	The trained MobileNetV2 model checks the numerical image and gives a probability for every class.
6. Choose highest result	NumPy argmax() selects the class having the highest probability.
7. Read class name	class_names.json converts the selected class number into a readable name such as Cotton_Leaf_Spot.
8. Show output	The probability is multiplied by 100 and displayed as confidence together with plant name, disease and status.

Easy flow to remember



Main code and faculty explanation

Only the most important points you need to remember

Load trained model	<code>model = tf.keras.models.load_model('leaf_disease_model.h5')</code>
Open and resize	<code>pil_img = Image.open(image_file).convert('RGB')</code> <code>pil_img = pil_img.resize((224, 224))</code>
Convert to numbers	<code>img_array = np.array(pil_img, dtype=np.float32)</code> <code>img_array = np.expand_dims(img_array, axis=0)</code>
Predict	<code>predictions = model.predict(img_array)</code>
Select highest class	<code>pred_idx = np.argmax(predictions[0])</code>
Get name and percentage	<code>raw_class = class_names[pred_idx]</code> <code>confidence = predictions[0][pred_idx] * 100</code>

Important correction

The JSON file does not predict. The trained model predicts first. After that, class_names.json changes the selected class number into a readable plant or disease name.

Example

Model probabilities	Class 0 = 0.05 Class 1 = 0.90 Class 2 = 0.05
Highest result	<code>argmax()</code> selects Class 1 because 0.90 is highest.
JSON class name	Class 1 = Cotton_Leaf_Spot
Final confidence	$0.90 \times 100 = 90\%$

Ready answer for faculty

"The user uploads or captures a leaf image. Django converts it into RGB format, resizes it to 224 by 224 and converts it into numbers. The trained MobileNetV2 model predicts probabilities for all classes. The system selects the highest probability, reads its disease name from the JSON class file, converts the probability into a percentage and shows the final result."